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B.Tech. Degree I Semester Examination December 2015

CE/EE/ME/SE 1103 A ENGINEERING MECHANICS (2015 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A (Answer ALL questions)

(10 × 2 = 20)

- I. (a) What are the divisions of force systems in a plane? Write their equilibrium equations.
 (b) State the laws of dry friction.
 (c) State any two applications of centroid in engineering.
 (d) How is product of inertia defined? What is its significance in engineering?
 (e) State the principle of virtual work. Explain it with an example.
 (f) State and explain D'Alembert's principle in rectilinear translation.
 (g) Define coefficient of restitution. What are its values for elastic, semi-elastic and plastic impacts?
 (h) Derive the equation for time period of small oscillations of simple harmonic motion choosing a suitable example.
 (i) State the principle of moment of momentum and the principle of angular momentum.
 (j) Derive the relationship connecting work and energy in translation or rotation, starting from the governing (kinetic) equation of motion.

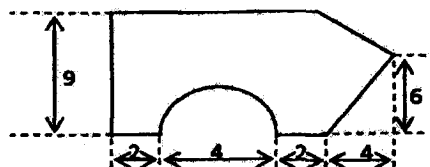
PART B

(4 × 10 = 40)

- II. (a) A roller of radius 60 cm weighing 600 N is to be pulled over a rectangular block of height 15 cm by a horizontal force applied at the end of a string wound around the circumference of the roller. Find the magnitude of the force, which will just turn the roller over the obstacle. (5)
 (b) A ladder 12 m long rests on a horizontal floor and leans against a vertical wall. The coefficient of friction between the wall and ladder is 0.15 and that between the floor and ladder is 0.25. Calculate the angle of inclination of the ladder with the floor when it is just about to slip. (5)

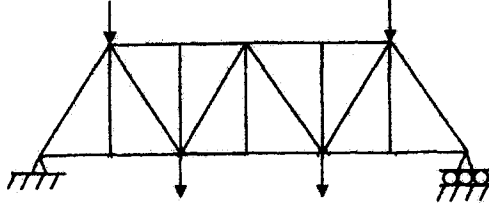
OR

- III. (a) Locate the centroid of the area shown in the figure. The curve in the lower middle is a semicircle. (6)



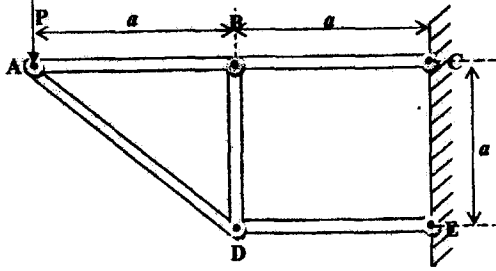
- (b) Calculate the second moment of area of an angle section with unequal legs, the length of the legs being 150 mm and 100 mm, thickness of the section 12 mm, with respect to the outer edge of the longer leg. (4)

- IV. Determine the forces in all the members of the plane truss, shown in the fig. The upper and lower chord members are of 2 m and vertical members are of 3 m length, each. The magnitudes of the forces are 5 kN , each. (10)



OR

- V. Determine the axial forces in the members BD, AD and DE of the frame shown in the figure. (10)



- VI. (a) A body moving with a uniform acceleration is observed to travel 33 m in the eighth second and 53 m in the 13^{th} second of its travel. Calculate its initial velocity and uniform acceleration. (4)

- (b) A train moves with a uniform speed of 36 kmph along a straight level track. At a certain instant, the traction is increased by 20% . What distance ' x ' will the train cover before acquiring a speed of 45 kmph if the resistance to motion is $1/200$ of the weight of the train? (6)

OR

- VII. (a) Two blocks A and B are released from rest on a 30° incline, when they are 18 m apart. The coefficient of friction under the upper block A is 0.2 and that under the lower block B is 0.4 . In what time will block A reach block B? After they touch and move as a single unit, what will be the contact force between them? The weights of the blocks A and B are 100 N and 80 N , respectively. (6)

- (b) When a ball of weight W rests on a spring of constant k , it produces a static deflection of 2.5 cm . How much will the same ball compress the spring if it is dropped from a height of 30 cm ? Neglect the mass of the spring. (4)

- VIII. (a) A car starts from rest on a curved road of radius 250 m and attains a speed of 18 kmph at the end of 60 seconds while travelling with a uniform acceleration. Find the tangential and normal accelerations of the car 30 seconds after it started. (4)

- (b) If the coefficient of friction, $\mu = 0.2$ between the wheel and the road, find at what speed a vehicle can get round a curve of 50 m radius without side-slip (6)
- on a level road.
 - on a road banked to a slope of 1 vertical to 8 horizontal.
 - At what speed can the vehicle travel on the banked road without any lateral frictional support?

OR

- IX. (a) A shaft of radius ' r ' rotates with constant angular speed ω in bearings for which the coefficient of friction is μ . Through what angle θ will it rotate after the driving torque is removed? (4)

- (b) A homogeneous plate having the shape of an equilateral triangle with edges of length ' a ' hangs in a vertical plane by pins at two of its vertices. Calculate the reaction at one of the pins, an instant after the other pin is removed. (6)